

# LI XIANG

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## 🎓 EDUCATION

**Yangtze University**, Jingzhou, China

Sep 2022 – Jun 2026

*B.Eng. in Software Engineering*

**GPA:** 3.11 / 4.0

**Relevant Coursework:** Discrete Mathematics, Probability & Statistics, Machine Learning, Algorithm Design & Analysis, Linear Algebra

## 🔬 RESEARCH EXPERIENCE

**GeoRouteNet: Geometry-Enhanced Neural Solver for Euclidean TSP** (*Undergraduate Thesis*) Nov 2025 – Jun 2026

Built upon reproduction and ablation studies of NAR4TSP, motivating a geometry-aware architecture for Euclidean TSP.

- Proposed **GeoRouteNet**, a non-autoregressive neural solver that injects geometric inductive biases into GNN node, edge, and attention representations.
- Designed a multi-candidate self-comparative reinforcement learning strategy to improve training stability and solution quality.
- Achieved **0.32% / 1.26% optimality gap on TSP50 / TSP100 (Beam-1000)**; on 27 TSPLIB EUC\_2D instances, reduced average gap from **17.12% to 3.60%** against the NAR4TSP baseline.
- Awarded **University-Level Outstanding Graduation Thesis**. Preprint available on arXiv:2606.22776.

## 🏆 COMPETITIVE PROGRAMMING & AWARDS

**University ACM Team Member**

Oct 2024 – Jun 2026

C++ Algorithm Training

- Solved 2000+ algorithmic problems across Codeforces, AtCoder, and LeetCode; focused on dynamic programming, graph algorithms, and combinatorics.

**National First Prize (Rank 219/2520)**, Mati Cup Programming Contest

Aug 2025

**National Second Prize**, RAICOM Robot Competition (CAIP Track)

Aug 2025

**National Second Prize**, Lanqiao Cup Programming Contest (C/C++)

Jun 2025

**Provincial Third Prize**, Group Programming Ladder Tournament (GPLT)

May 2025

**Top 15%**, Programming Ability Test (PAT Advanced)

Sep 2025

## </> SELECTED PROJECTS

**poetry-GPT: Decoder-only Transformer for Chinese Poetry Generation** May 2025 – Jun 2025

*PyTorch, CUDA* Independent Project

- Built a 7-layer decoder-only Transformer (~14M parameters) in PyTorch for classical Chinese poetry generation.
- Developed a character-level preprocessing pipeline over 2.2M Chinese characters and implemented temperature, top-k, and nucleus sampling for autoregressive decoding.

## 📖 SKILLS & LANGUAGES

- **Programming:** Python, C/C++, Java, SQL
- **Research & ML:** Literature reading and paper reproduction; proficient in implementing deep learning models in PyTorch; familiar with reinforcement learning concepts
- **Algorithms:** Strong foundation in data structures and algorithms, with extensive practice in dynamic programming, graph algorithms, and combinatorics
- **Languages:** Chinese (Native), English (CET-6)